

T0-Theory: A Unified Physics from a Single Number Comprehensive Summary of the Document Collection

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Abstract

The T0-Theory (Time-Mass Duality) represents a fundamental paradigm shift in theoretical physics. In simple words: Imagine the universe as a large puzzle in which everything—from the smallest particles to the vast cosmos—fits perfectly together, without loose ends. The central result of this work is the insight that **all natural constants and physical parameters can be derived from a single dimensionless number**: the universal geometric constant

$$\xi \approx \frac{4}{3} \times 10^{-4},$$

where $\xi \approx 4/3 \times 10^{-4}$. Imagine ξ as the "master key" of the universe—a tiny number that emerges from the basic form of three-dimensional space and unlocks explanations for gravity, the speed of light, particle masses, and more.

This collection of over 200 scientific documents systematically develops a complete physical theory that unifies quantum mechanics, relativity, and cosmology—based on the principle of absolute

time T_0 and the intrinsic time-field-mass relationship. In everyday language: It's as if we are rewriting the rules of physics so that time is stable and reliable (not flexible as in Einstein's view), while mass can change like sand in the wind, all connected through this elegant geometric idea.

The fundamental documents follow a purely geometric path, deriving ξ from the three-dimensional structure of space and constructing all other constants from it, including the fine structure constant

$$\alpha \approx 1/137,$$

where $\alpha \approx 1/137$, particle masses, and coupling strengths, without introducing additional free parameters. No more arbitrary numbers; everything flows from a single simple source, making the universe less random and more like a beautifully designed whole. Remarkably, the theory postulates a static universe without expansion, as detailed in the [CMB document](#) (verified link exists), thereby rendering concepts like dark matter or dark energy superfluous.

This book presents the current state of the T0 Time-Mass Duality Framework and its applications to particle masses, fundamental constants, quantum mechanics, gravity, and cosmology.

The main part of the book consists of a series of core T0 documents. These chapters reflect the current understanding of the theory and its quantitative consequences. Wherever possible, the material has been reorganized and unified to make the structure of the theory as transparent as possible.

The "Live" version of the theory is maintained in a public GitHub repository:

<https://github.com/jpascher/T0-Time-Mass-Duality>

The $\text{\LaTeX}$ sources of the chapters in this book come from this repository. If conceptual or numerical errors are found, they will be corrected there first. This means that the PDF version of the book you are reading is a snapshot of a continuously evolving project. For the most current version of the documents, including new appendices or corrections, the GitHub repository should always be considered the primary reference.

The intention of this compilation is twofold:

- to provide a coherent, readable path through the core ideas and results of the T0-Framework;

- to document the historical development of these ideas in the appendix, including false starts, interim formulations, and early adjustments to experimental data.

Readers who are primarily interested in the current formulation of the theory can focus on the core chapters. Readers who are also interested in the considerations and trial-and-error process behind the theory are invited to study the appendix material in parallel.

1 The Core Principle: Everything from One Number

The fundamental insight of the T0-Theory can be summarized in one sentence:

Key Result

Central Theorem of the T0-Theory: All physical constants—gravitational constant G , Planck constant $\hbar$, speed of light c , elementary charge e , as well as all particle masses and coupling constants—can be mathematically derived from a single dimensionless number: the universal geometric constant

$$\xi = \frac{4}{3} \times 10^{-4},$$

which emerges from the fundamental three-dimensional space geometry via

$$\xi = \frac{4\pi}{3} \cdot \frac{1}{4\pi \times 10^4}.$$

From ξ follows the fine structure constant as:

$$\alpha = f_\alpha(\xi) \approx \frac{1}{137.035999084},$$

where α serves as a secondary electromagnetic coupling without primacy.

In everyday language, this means: We have reduced the “why” of physics to a single, space-born number—no magic, just geometry doing the heavy lifting.

.2 Foundations of the T0-Theory

Time-Mass Duality

In contrast to standard physics, where time is relative and mass is constant, the T0-Theory postulates:

- **Absolute Time Measure** T_0 : Time flows uniformly everywhere in the universe—like a universal clock that ticks the same for everyone, no matter where you are.
- **Variable Mass**: Mass varies with the energy content of the vacuum—imagine mass as flexible, changing depending on the “hum” of the empty space around it.
- **Intrinsic Time Field** $T(x, t)$: Every particle carries its own time field—each building block of matter has its personal timer that influences its behavior.

The fundamental relationship is:

$$m(x) = \frac{\hbar}{c^2 T(x, t)(x)} = m_0 \cdot (1 + \kappa \Phi(x)),$$

where κ is traceable back to ξ via geometric scaling. Mathematically, this duality treats time and mass as variables, ensuring that the framework remains fully compatible with established mathematical structures while enabling a unified description of physical phenomena. Simply put: By letting time and mass dance as adaptable partners, we keep the mathematics clean and intuitive, connecting old ideas with new ones without breaking a sweat.

The Parameter ξ

The central parameter of the theory is:

$$\xi = \frac{4}{3} \times 10^{-4},$$

a purely geometric construct from 3D space that connects quantum mechanics with gravity. This parameter encodes the fundamental coupling between energy and spatial structure, from which all hierarchies emerge. It is like the ratio that tells space how to “scale” energy—small but powerful, whispering the secrets of why electrons are light and protons heavy.

.3 Derivation of All Natural Constants

Everything Follows from ξ

The T0-Theory demonstrates that:

1. Gravitational Constant:

$$G = f_G(\xi, m_P, c, \hbar),$$

where all inputs are reducible to ξ -scaled geometric units. Gravity? Just a wave from the geometry of space, tuned by ξ .

2. **Particle Masses** (Electron, Muon, Tau, Quarks): Particle masses follow a universal scaling law analogous to the ordering principles of atomic energy levels, where quantum numbers (n, l, j) dictate hierarchical structures in a manner similar to atomic shells and subshells—imagine particles stacked like floors in a building, each level set by simple rules, similar to how electrons orbit atoms. Thus,

$$\frac{m_e}{m_P} = g(\xi), \quad \frac{m_\mu}{m_e} = h(\xi), \quad \frac{m_\tau}{m_\mu} = k(\xi),$$

via universal scaling laws $\xi_i = \xi \times f(n_i, l_i, j_i)$. No more guessing why some particles are 200 times heavier; it's all patterned like a cosmic family tree.

3. **Coupling Constants** (Electroweak, Strong, Electromagnetic):

$$\alpha_W = f_W(\xi), \quad \alpha_s = f_s(\xi), \quad \alpha = f_\alpha(\xi).$$

These “strengths” of forces? Derived like branches from the same geometric trunk.

4. **Cosmological Parameters**: Static universe metrics and CMB temperature $T_{\text{CMB}} = f_{\text{CMB}}(\xi)$, with redshift mechanisms derived from time-field variations (see [CMB document](#) for detailed explanation without expansion).

.4 Experimental Predictions

The T0-Theory makes precise, testable predictions:

Foundation

Concrete Predictions:

- **Anomalous Magnetic Moment:** $(g - 2)_\mu$ calculation solely from ξ —a quirky electron-like wobble explained without extras.
- **Koide Formula:** Exact mass relation of leptons via ξ -scaling—the mathematics that connects the weights of three particles in a clean loop.
- **Redshift:** Modified interpretation without expansion, controlled by ξ —why distant stars appear “stretched” without the universe inflating.
- **CMB Anisotropies:** Explanation through time-field variations rooted in ξ —the microwave “echo” of the cosmos as geometric echoes.

These are not wild guesses; they are verifiable with today's laboratories and invite everyone—physicists or curious minds—to put the theory to the test.

.5 Structure of the Document Collection

This collection includes:

- **Foundations:** Mathematical formulation of time-mass duality under ξ -geometry—the basics explained step by step.
- **Quantum Mechanics:** Deterministic interpretation, Bell inequalities—quantum madness made predictable and local.
- **Quantum Field Theory:** Lagrangian formalism in the T0-Framework—fields dancing to a unified melody.
- **Cosmology:** Static universe, redshift, CMB—a stable universe that still surprises, without expansion, dark matter, or dark energy.
- **Particle Physics:** Mass spectrum, anomalous moments, Koide formula—the particle zoo tamed.
- **Technical Applications:** Photon chip, RSA cryptography—real tricks from the theory.

- **Experimental Tests:** Verifiable predictions—tangible ways to investigate the ideas.

Note: The documents consistently follow the geometric ξ -path, deriving all physics from 3D space principles, with α and other constants appearing as emergent features. We have woven simple language throughout so that non-experts can dive in without drowning in jargon.

.6 Introduction: The Milestone of Vibrations

The foundation of my T0-Theory did not arise from abstract equations, but from practical work in communications engineering, acoustics, and music theory. Long before I could consider empty space as a dynamic field, I was engaged with vibrations in concrete bodies—for example, the accordion reed [3]. This small, vibrating membrane in an accordion produces sound through resonance in the “empty” air space between: Frequency and amplitude interact dually, without the space remaining “empty.” It was a milestone: Here I saw emergence pure—vibration (time) and medium (space) create harmony, without singularities.

This unbiasedness—why not see ϵ and μ in QM and EM as dual resonators?—later led to the vacuum approach. In natural units ($\hbar = c = 1$), setting α to 1, and everything clicks: EM constants become geometric, QM/RT unified. The warning against “translation” ($\epsilon_0 \neq \mu_0$ naively) was crucial—in T0, ξ “modulates” both without loss. From acoustics (resonances in cavities) and communications engineering (Fourier dualities time-frequency [4]) came the entry: Empty space as a resonant vacuum, carried by EM constants ($\epsilon_0, \mu_0, c = 1/\sqrt{\epsilon_0\mu_0}$). Music theory reinforced it: Harmonies (Pythagorean 3:4:5 tetrahedra) as fractal overtones hinting at tetra networks.

.7 The Vacuum Approach: From Acoustics to Duality

From acoustics (resonances in cavities) and communications engineering (Fourier dualities time-frequency [4]) came the entry: Empty space as a resonant vacuum, carried by EM constants ($\epsilon_0, \mu_0,$

$c = 1/\sqrt{\epsilon_0\mu_0}$). Music theory reinforced it: Harmonies (Pythagorean 3:4:5 tetrahedra) as fractal overtones hinting at tetra networks.

T0 formalizes it: The duality $T_{\text{field}} \cdot E_{\text{field}} = 1$ connects time (vibration) and energy (mass), with ξ as the geometric seed. In natural units, set $\alpha = 1$: The Coulomb potential $V(r) = -1/r$ becomes purely geometric, the Bohr radius $a_0 = 1$ a unit length. Tetrahedral networks “cover” the time field—emergence of charge/mass without point singularities.

The derivation of α :

$$\alpha = \xi \cdot \left(\frac{E_0}{1 \text{ MeV}} \right)^2, \quad E_0 = 7,400 \text{ MeV}, \quad (1)$$

yields $\approx 1/137$ [2], corrected by fractal steps $\prod_{n=1}^{137} (1 + \delta_n \cdot \xi \cdot (4/3)^{n-1})$ to CODATA precision. No “translation trap”—SI conversion via $S_{T0} = 1.782662 \times 10^{-30} \text{ kg}$ projects geometry into the measurement world. Setting $\alpha = 1$ in natural units ($\hbar = c = 1$) makes sense: It reduces EM fluctuations to pure resonance, like in the accordion reed [3]—vacuum as an acoustic medium where ϵ_0 and μ_0 resonate dually, without naive exchange.

This approach was unbiased: If you set $c = 1$, why not α ? The consequence: Tetrahedral networks emerge naturally to “cover” the time field, and fractal iterations (137 steps) stabilize the emergence of charge and mass. It clicks because physics is dimensionless patterns—from the tangible (vibrations) to the abstract (vacuum).

.8 Convergence with Synergetics: Independent Paths

Despite a different approach, T0 and Synergetics converge: Bucky Fuller’s tetrahedron as the “minimum structural system” [1] (closest-packing spheres) fractions to vector equilibria—exactly like T0’s networks “pack” the vacuum. The 137-frequency tetrahedron (2,571,216 vectors = $137 \times 9,384 \times 2$) mirrors T0’s renormalization: Proton-MeV (938.4) as an emergent ratio.

The independence is the highlight: From acoustic resonances (accordion reed as vacuum prototype [3]) to duality, without Fuller—yet it “clicks” at $\alpha = 1$. Synergetics provides the “foundation” that you intuitively supplemented: Tetra-fractionation stabilizes vortices (charge), 137 steps as spin transformations (tetra $\rightarrow$ octa $\rightarrow$

icosa). The long-term engagement with vibrations (accordion reed as resonance milestone) and unbiasedness (ϵ_0 and μ_0 as dual resonators, without naive translation) independently led to vacuum duality.

Approach	T0 (Vacuum Duality)	Synergetics (Tetra-Fraction)
Entry	Acoustics/Resonance in empty space	Closest-Packing Spheres
α -Derivation	$\xi \cdot (E_0)^2$ (nat. units: $\alpha = 1$)	137-Frequency Vectors
Time Field	Tetra networks cover duality	Morphological Relativity
Emergence	Charge as vortex (finite U)	Vector-Tensor Intertransformation
ϵ_0/μ_0	Dual Resonators (modulated via ξ)	Tensor Forces in Packing

Table 1: Convergences: T0 and Synergetics—extended by duality elements

The convergence is no coincidence: Both reduce to tetrahedral patterns, but T0 from vacuum resonance (accordion reed as prototype [3]), Synergetics from packing [1]. Setting $\alpha = 1$ in natural units (Coulomb $V(r) = -1/r$, Bohr radius $a_0 = 1$) shows: It “makes sense” because empty space is geometric— ϵ_0 and μ_0 as dual “modulators,” without translation traps.

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